

Translated

From: e-maxx.ru

Discrete Logarithm

The discrete logarithm is an integer x satisfying the equation

$$a^x \equiv b \pmod{m}$$

for given integers a , b and m .

The discrete logarithm does not always exist, for instance there is no solution to $2^x \equiv 3 \pmod{7}$. There is no simple condition to determine if the discrete logarithm exists.

In this article, we describe the **Baby-step giant-step** algorithm, an algorithm to compute the discrete logarithm proposed by Shanks in 1971, which has the time complexity $O(\sqrt{m})$. This is a **meet-in-the-middle** algorithm because it uses the technique of separating tasks in half.

Algorithm

Consider the equation:

$$a^x \equiv b \pmod{m},$$

where a and m are relatively prime.

Let $x = np - q$, where n is some pre-selected constant (we will describe how to select n later). p is known as **giant step**, since increasing it by one increases x by n . Similarly, q is known as **baby step**.

Obviously, any number x in the interval $[0; m)$ can be represented in this form, where $p \in [1; \lceil \frac{m}{n} \rceil]$ and $q \in [0; n]$.

Then, the equation becomes:

$$a^{np - q} \equiv b \pmod{m}.$$

Using the fact that a and m are relatively prime, we obtain:

$$a^{np} \equiv ba^q \pmod{m}$$

This new equation can be rewritten in a simplified form: